Example Ocean Account Tables

Coastal District Example -- Illustrative data for training purposes

Table 1: Ecosystem Extent Account (2018 to 2023)

	Coral reefs (M1.3)	Seagrass (M1.1)	Mangroves (MFT1.2)	Other	Total
Opening extent (ha)	**1,240**	**860**	**315**	**7,585**	**10,000**
Additions: natural expansion	30	20	35	85	170
Additions: managed restoration	10	15	5	0	30
Additions: reclassification	0	0	0	70	70
Total additions	**40**	**35**	**40**	**155**	**270**
Total reductions	**100**	**70**	**15**	**85**	**270**
Net change	**-60**	**-35**	**+25**	**+70**	**0**
Closing extent (ha)	**1,180**	**825**	**340**	**7,655**	**10,000**

Total accounting area is conserved (10,000 ha). Net change across all types sums to zero.

Table 2: Ecosystem Type Change Matrix (hectares)

Opening \ Closing	Coral	Seagrass	Mangrove	Other	Opening total
Coral reefs	1,140	15	0	85	**1,240**
Seagrass	10	790	5	55	**860**
Mangroves	0	0	300	15	**315**
Other	30	20	35	7,500	**7,585**
Closing total	**1,180**	**825**	**340**	**7,655**	**10,000**

Green cells = stable area (no change). Off-diagonal = transitions between ecosystem types.

Table 1b: Ecosystem Extent Time Series (multiple accounting periods)

Each period's closing becomes the next period's opening. This table builds up over time.

Ecosystem type	2010 (ha)	2014 (ha)	2018 (ha)	2023 (ha)	Total change	Change (%)
Coral reefs (M1.3)	1,420	1,340	1,240	1,180	**-240**	**-16.9%**
Seagrass (M1.1)	950	910	860	825	**-125**	**-13.2%**
Mangroves (MFT1.2)	280	290	315	340	**+60**	**+21.4%**
Other	7,350	7,460	7,585	7,655	**+305**	**+4.1%**

Total	**10,000**	**10,000**	**10,000**	**10,000**	**0**	**0%**

Net change per accounting period

Ecosystem	2010-2014	2014-2018	2018-2023	Trend
Coral reefs	-80	-100	-60	Declining (slowing)
Seagrass	-40	-50	-35	Declining (slowing)
Mangroves	+10	+25	+25	Increasing (stable rate)

Reading the time series: Coral and seagrass losses are slowing across periods, suggesting either reduced pressures or approaching a degradation floor. Mangrove gains are consistent, likely reflecting sustained restoration. The time series is where the value of repeated accounting becomes visible: a single snapshot shows status, but the series shows trajectory.

Example Ocean Account Tables (continued)

Table 3: Ecosystem Condition Account -- Coral Reefs (M1.3)

Marine Reserve Example, 2019 to 2024. CI scale: 0 = degraded, 1 = reference condition.

Indicator	Unit	Direction	Reference	Opening value	Opening CI	Closing value	Closing CI	Change	Status
Live coral cover	%	Higher better	50%	31.1	0.62	26.6	0.53	-0.09	Declining
Macroalgae cover	%	Higher worse	60%	22.6	0.62	27.4	0.54	-0.08	Declining
Fish biomass	kg/ha	Higher better	500	213.8	0.43	193.8	0.39	-0.04	Declining
Recruit density	per m2	Higher better	15	8.5	0.57	6.2	0.41	-0.15	Declining
Reef relief	cm	Higher better	150	82.5	0.55	78.0	0.52	-0.03	Stable
Bleaching prevalence	% colonies	Higher worse	100%	8.5	0.92	18.3	0.82	-0.10	Declining
Composite CI					**0.62**		**0.53**	**-0.08**	**Declining**

Reading this table: The composite condition index declined from 0.62 to 0.53 over 5 years. The reef is at roughly half its reference condition, with coral cover, fish biomass, and bleaching showing the largest declines. Reef relief (physical structure) remained relatively stable.

Example Ocean Account Tables (continued)

Table 4a: Ecosystem Service Physical Supply (Part A)

Part A is a complete account on its own. No monetary data needed.

Service	Category	Unit	Coral reefs	Seagrass	Mangroves	Total
Fish provisioning	Provisioning	kg/yr	120,000	45,000	15,000	**180,000**
Wood provisioning	Provisioning	tonnes/yr			85	**85**
Carbon sequestration	Regulating	Mg CO2/yr		1,040	1,295	**2,335**
Coastal protection	Regulating	m coastline	12,000		3,500	**15,500**
Nursery habitat	Regulating	kg biomass/yr	4,500	2,800		**7,300**
Sediment retention	Regulating	m3 CaCO3/yr	8,400			**8,400**
Recreation (reef)	Cultural	visitors/yr	15,000			**15,000**
Recreation (mangrove)	Cultural	trips/yr			2,400	**2,400**
Gleaning	Cultural	hours/yr		18,000		**18,000**

Table 4b: Ecosystem Service Monetary Supply (Part B) -- USD/yr

Service	Method	Type	Coral reefs	Seagrass	Mangroves	Total
Fish provisioning	Resource rent	Market	140,000	52,500	17,500	**210,000**
Wood provisioning	Substitute cost	Non-market			12,750	**12,750**
Carbon sequestration	SCC ($51/Mg)	Non-market		53,040	66,045	**119,085**
Coastal protection	Replacement cost	Non-market	1,200,000		175,000	**1,375,000**
Nursery habitat	Productivity change	Market	78,750	49,000		**127,750**
Sediment retention	Replacement cost	Non-market	420,000			**420,000**
Recreation	Direct expenditure	Market	1,275,000		84,000	**1,359,000**
Gleaning	Equiv. wage + market	Mixed		63,000		**63,000**
TOTAL			**3,113,750**	**217,540**	**355,295**	**3,686,585**

Key insight: Regulating and cultural services (coastal protection $1.4M, recreation $1.4M) account for 75% of total ecosystem service value. Fish provisioning, often assumed to be the dominant service, represents only 6% of the total. This pattern is common in tropical coastal

systems and has direct implications for policy: protecting reef structure and water quality protects the highest-value services.